Microsoft

GH-900

GitHub Foundations

[GitHub Foundations Part 2 of 2]

Module 9

Contribute to an open-source project on GitHub

Agenda

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GitHub Foundations Part 2

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Introduction

Discover how to create pull requests and communicate with project maintainers effectively to get your changes accepted. Learn about the benefits of getting involved with open-source communities.

By the end of this module, you'll be able to:

- Find open-source projects and tasks to contribute to in GitHub.
- Create pull requests to open-source projects.
- Implement best practices to communicate with open-source maintainers and perform code reviews.
- Find and engage with open-source communities.

How to create a project

Most projects will have these documents at the top level of the repository:

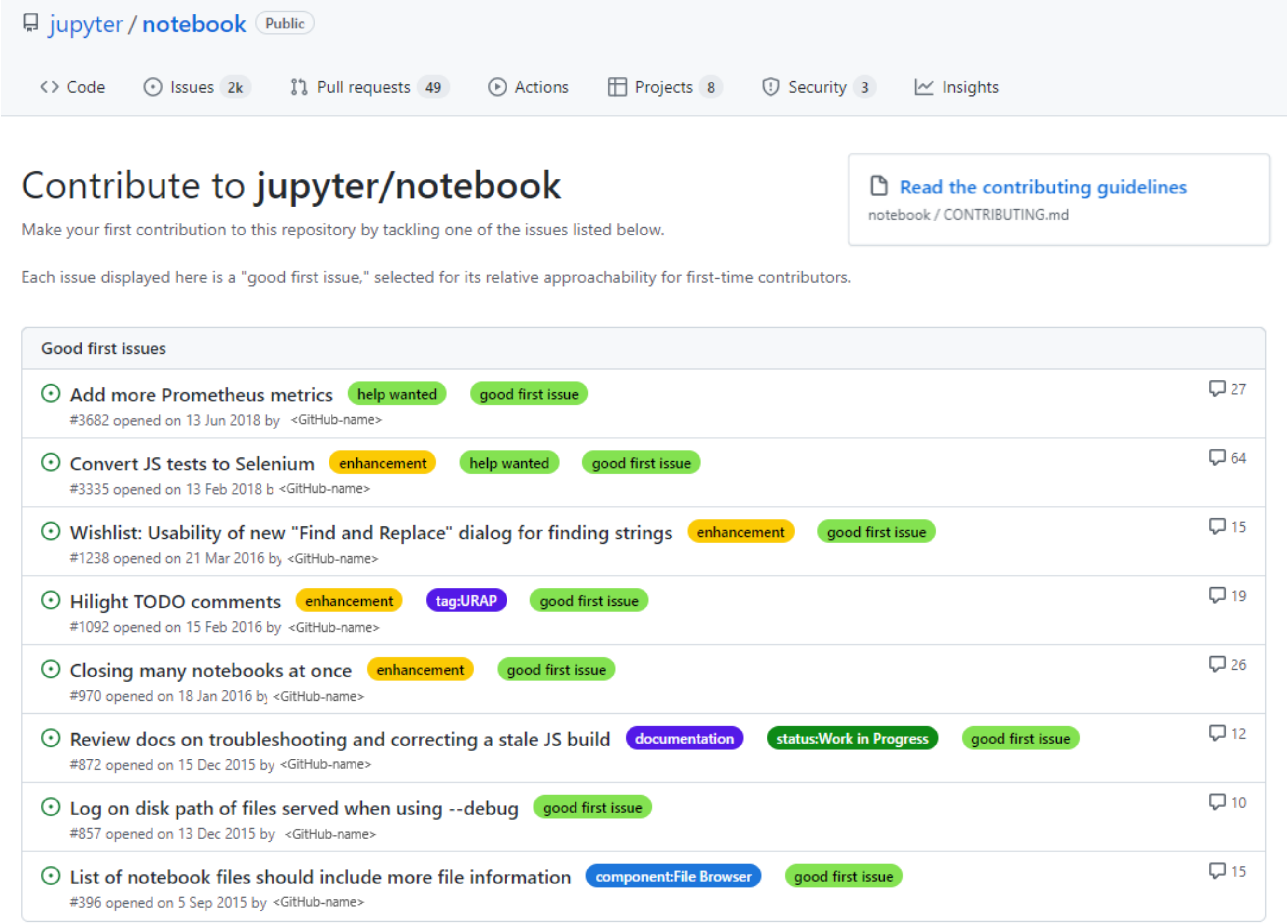
- **LICENSE:** The project must contain an open-source license. If the project doesn't have a license, it's not open source.
- **README:** File that serves as the welcome page for the project. It generally provides information on how to get started using the project. It's also common for it to add information on how to engage with the community.
- **CONTRIBUTING:** This document provides guidance on how to contribute to the project. It usually describes how the contribution process works, and includes details on how to set up your development environment.
- **CODE_OF_CONDUCT:** Sets ground rules for community members. By doing so, it helps make the community a safe and welcoming environment for all.

How to create a project (cont.)

Perhaps you've already identified something to work on, such as fixing broken links or updating the docs.

A good way to find issues to help with is by visiting the project's **/contribute** URL.

For example: https://github.com/jupyter/notebook/contribute .



Contribute to an open-source repository

A pull request includes the following activities:

GH $\xrightarrow{\text{Fork}}$ GH

PR

Local

1) **Fork the Repository**
Select the Fork button on the GitHub project page to create a copy of the repository on your GitHub account.

2) **Clone the Repository**
Copy the repository URL and use the git clone <REPOSITORY_URL> command in your terminal to create a local copy.

3) **Create a New Branch (Optional)**
Use git checkout -b <BRANCH_NAME> to create a new branch for your changes, allowing for separate contributions.

Make and Commit Changes
Make the desired changes, then use git add . and git commit -m "<COMMIT_MESSAGE>" to stage and commit them.

5) **Push Changes to GitHub**
Push your changes to the remote repository with git push --set-upstream origin <BRANCH_NAME>.

6) **Create a Pull Request**
Open your project fork on GitHub, select the Compare & pull request button, fill in the title and description, and create the pull request.

Exercise - Create your first pull request

Using the [First Contributions](#) project, you practice forking, cloning, and submitting a pull request.

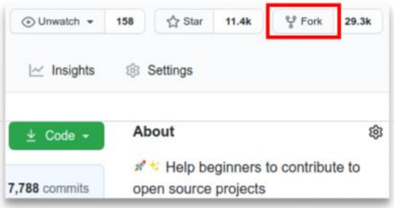
It has guides for both using the command line and several graphical user interfaces (GUIs).

First Contributions

This project aims to simplify and guide the way beginners make their first contribution. If you are looking to make your first contribution, follow the steps below.

If you're not comfortable with command line, [here are tutorials using GUI tools.](#)

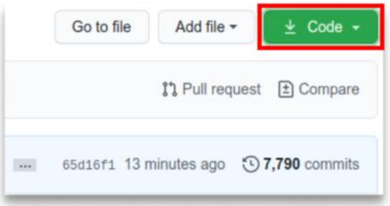
If you don't have git on your machine, [install it](#).



Fork this repository

Fork this repository by clicking on the fork button on the top of this page. This will create a copy of this repository in your account.

Clone the repository



Now clone the forked repository to your machine. Go to your GitHub account, open the forked repository, click on the code button, then on SSH tab and then click the *copy to clipboard* icon.

Open a terminal and run the following git command:

```
git clone "url you just copied"
```

Start course

Next Steps

Getting Involved in the Community

- Find contributors in the comment sections of issues and pull requests, or through the **Contributors** section under **Insights**.
- Follow organizations and enterprises on GitHub to stay updated on their public activity using your personal dashboard.
- Attend meetups, conferences, and explore archives of past events, podcasts, newsletters, and mailing lists.

Code Reusability

- Publish reusable code as a stand-alone library, mirror the project with added functionality, or create a **GitHub Action**.
- GitHub Actions automate tasks in workflows and can be submitted to the GitHub Marketplace for discoverability.
- Consider the responsibilities of maintaining a project, including handling feedback, updating documentation, and managing issues.

Maintaining and Sharing Code

- Assess your capacity to manage a project if it gains popularity, including involving more people to share the workload.
- Set clear expectations in your project's README file regarding your availability and support.
- Share your code through public gists or blog posts if you prefer not to host it on GitHub.

Summary

In this module, you wanted to take part in the open-source community by learning how to provide support and contribute to open-source projects using GitHub.

You learned how to:

- Find projects to contribute to on GitHub.
- Familiarize yourself with the project and its guidelines.
- Use the issue tracker and labels to find tasks to work on.
- Use GitHub Sponsors to financially support your favorite projects and open-source contributors.

End of presentation